COMPARISON OF STEREO IMAGE DESCRIPTORS FOR RECORDED MUSIC

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ABSTRACT

The stereo image (i.e. the distribution of sound between left and right channels) is an important feature to extract from a piece of recorded music, with downstream uses such as classification, quality assessment, and listening enhancement. While several stereo image descriptors exist in the literature, there is currently no comparison between them and no studies to suggest which one is most suitable for any particular task. This paper compares 4 such descriptors, finds synchronicities between them, and compares their results. Audio signals designed with the intention of exposing each descriptor's idiosyncrasies and biases are constructed, and the resulting stereo image descriptions analysed. Furthermore, examples of commercial music are given to demonstrate the effectiveness of each method in detecting particular audio features. Results suggest that none of the analysed descriptors can indicate all facets of a stereo image, so a combination should be used in further work.

1. INTRODUCTION

The vast majority of recorded music available today was recorded and produced in stereophonic format. The musicians and producers involved will often pay close attention to the placement of sound objects in the stereo field, and various pieces of equipment such as goniometers, phase scopes, and stereo balance indicators have been standard equipment in recording studios since the inception of stereo sound. There is therefore a wealth of spatial information encoded in the final recording. Extracting and disentangling that information has been the subject of research for many years, in various different guises.

While there has been a large amount of research into spatial information in binaural audio (driven by a particular interest in the human auditory system) [1], stereo music signals often exhibit spatial features that are not possible in binaural audio, and therefore they require tailored extraction techniques [2]. This is particularly prevalent in recent years because of the near ubiquity of headphone listening [3] - a trend that was not necessarily predictable in previous decades when a lot of music was produced. Importantly, there are numerous recording and production techniques that affect the spatial feeling of a recording [4], each resulting in different degrees of 'headphone-friendliness'.

While there is an entire research field dedicated to determining *spatial quality* (i.e. measuring spatial degradation between a reference and manipulated signal) the standard methods (e.g. [5]) do not provide an overall *spatial description*, and are anyway routed in binaural analysis. Luckily, several examples of spatial analysis directly on stereo audio signals exist in the literature, but as yet there has not been a comparison of them or a discussion of their effectiveness. The aim of this paper is to provide such discussion. It will begin by summarizing the relevant literature in section 2, describing and drawing links between existing objective spatial descriptors. It will then demonstrate the abilities and idiosyncrasies of each one by applying them to audio signals constructed especially to test the limits of the algorithms. Furthermore, it will present examples of real music chosen for their interesting uses of stereo field, and discuss the effectiveness of each algorithm in detecting those musical features. These signals are detailed in sections 3.1 and 3.2 respectively. The results of each descriptor are shown and discussed in section 4. Additional discussion and concluding remarks are offered in sections 5 and 6.

2. PREVIOUS WORK

Measures of spaciousness in recorded music come in several flavours. There are perceptual qualities which aim to describe how a listener will experience the reproduction of a recording using a few scalar values. [6] specifies a long list of spatial attributes perceived by listeners, but three major ones often referred to by other authors are 'ensemble source width' (sometimes called 'apparent source width'), 'extent of immersion' (sometimes called 'listener envelopment'), and 'extent of reverberation' [7, 8, 2]. [9] and [10] define a scalar measurement of 'spaciousness' (which is very closely related to source width).

The above perceptual qualities are often inferred using objective measurements, which aim to describe the physical properties of an audio signal (usually in higher dimensionality) than the perceptual scalars. [4, 11] are seminal works describing the 'panning index' – a description of amplitude panning for each time-frequency bin in a stereo spectrogram. [4] also presents the 'ambience index' – a measure of inter-channel correlation in time-frequency (TF) bins. While they target re-synthesis and upmix as use cases, the underlying techniques have become the basis for many stereo image descriptors since. [12] is another seminal work which describes a frequency-azimuth plane constructed from phase cancellation of different frequency components. Again, it targets re-panning and re-synthesis by assigning time-frequency bins to individual sources rather than stereo image description. While not specifically the subject of this paper, for a comparison of similar techniques for directional segmentation and re-synthesis, the reader is referred to [13]. [14, 15] reuse the panning index from [4] (renamed to 'Stereo Panning Spectrum (SPS)') and derive a set of Stereo Panning Features (SPF) that are used for genre classification and Music Information Retrieval (MIR). Similarly, [16] de-

* This work was supported by the KKS Foundation

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tails a method to produce stereo image descriptors centred around a histogram of spatial energy, but uses the frequency-azimuth plane from [12] as its basis. Another histogram method, this time based on the SPS from [14] is described in [17, 18], citing measurement of music similarity as the primary use case. In a similar vein, and citing classification as a use case, [19] combines cochleagram differences with an analysis of phase differences between channels to form two feature sets - the 'Amplitude Stereo Features (ASF)' and 'Phase Stereo Features (PSF)'.

[20, 21, 22] collectively define a new paradigm of classifying stereo field whereby stereo mixtures are modelled as combinations of centre, panned, and uncorrelated sources; Although, they specifically state that most music signals will fall into a class that is very difficult to identify and is therefore omitted from the model. [23] uses the Interaural Intensity Difference (IID) and Interaural Phase Difference (IPD) of an audio signal to derive bin-level measurements of 'panning position' and 'panning clarity', which are subsequently used for sonogram visualization and manipulation. Similar to the phase-based features defined in [19], there are a group of algorithms based on the emulation of a studio goniometer - [24, 25] use both a goniometer box-counting technique and inter-channel correlation analysis to derive a time-varying measurement of spaciousness. Interestingly though, [26] found that subjective impressions of spaciousness in various musical styles were largely contradictory to the results of goniometer analysis.

Directional Audio Coding (DirAC) is a technique that converts ambisonic audio to a representation of directionality and diffuseness [27]. While it can be used on stereo audio, the conversion from stereo to ambisonic assumes the use of speakers rather than headphones [28].

More recently a Machine Learning (ML)-based technique for disentangling spatial information has been demonstrated with a use case of upmixing stereo to multichannel [29]; The input to the system is a combination of multichannel audio and it's associated stereo downmix, which is then re-upmixed by a neural network. A latent vector formed by running multichannel audio through one half of the network is demonstrated to represent the spatial information contained in the original signal.

2.1. Selected algorithms

This paper will focus on 4 objective measurements, that a) are high-dimensional, b) are specifically stated to be used for MIR, and c) do not require training on a dataset. Specifically, two forms of Panning Histogram (PH) – [16] and [17], and two forms of summary features – SPF from [14] and ASF/PSF from [19]. For mathematical definitions the reader is referred to the original papers, but a summary of each is given here.

2.1.1. Stereo Panning Features — Tzanetakis, 2007

The left and right channels are converted into complex TF regions by chunking, windowing, and the Short-time Fourier Transform (STFT). These two TF regions are converted into a panning spectrogram, which is then split into high, mid, and low frequency bands. Each band plus the full spectrum are collapsed by calculating the Root Mean Squared (RMS) across the frequency dimension, and smoothed in time by calculating a running mean and standard deviation over M frames. Those time vectors and then collapsed into a single value by taking the mean and standard deviation over time. The result is a 16-dimensional vector.

2.1.2. Panning Histogram — Sarroff, 2008

The left and right channels are split into chunks and transformed into the frequency domain using a Fast Fourier Transform (FFT). Each FFT frame is converted into a frequency-azimuth (FA) plane by calculating phase cancellations at various time offsets. The left and right FA planes are concatenated, and phase nulls are converted to peaks. The output histogram is then calculated across all frames and all frequencies.

Note: The paper continues to collapse the histogram into a single scalar representing overall width, but we will not consider that here since the intent is to analyse the high-dimensional histogram representation itself.

2.1.3. Panning Histogram — Gomez, 2008

The left and right channels are split into chunks and transformed into magnitude spectrograms using a STFT. The ratio of power spectra is calculated for each frame, resulting in a panning factor in the range ±45° for each frequency bin. These values are warped by a function that accounts for the azimuthal regions where humans perceive location more or less accurately. Each frame is then converted to an energy-weighted histogram and the mean over time is taken. The result is a histogram of panning energy at each azimuth bin.

Note: The paper continues to collapse the histogram into a shorter vector for classification purposes. For the same reasons as above we will not consider that step here.

2.1.4. Amplitude/Phase Panning Features — Tardieu, 2011

The ASF are created from a stereo cochleagram. Left and right audio channels are split into frequency sub-bands using gammatone filters. The amplitude difference of the two cochleagram channels is taken to produce a Cochleagram Difference (CD). The ASFs are created by taking the mean and standard deviation over various axes of the CD. The mathematical definition of a cochleagram is not given in [19], so for this paper we have followed the steps in [30].

The PSF are created separately. Left and right channels are first transformed into a complex TF region via a STFT. For each TF bin, the phase component and bin centre frequency is used to create a stereo cosine wave. The ratio of de-phasing is then calculated between channels, generating a Spectral Stereo Phase Spread (SSPS) spectrogram. The frequency bins are then grouped into the same frequency bands as the cochleagram and weighted by the energy in those bands. Summary features are produced by taking the mean and standard deviation over various axes. Unfortunately the PSF results shown in the original paper were difficult to reproduce for this analysis, and they were therefore omitted from the comparison. See section 4.3.1 for a detailed explanation.

3. METHOD

We probe the inner workings of each spatial descriptor in two ways. Firstly we construct some artificial signals that are designed to test the algorithms, and examine the output generated by each one. Secondly, we select some excerpts of commercially released music that exhibit specific spatial qualities and again examine the outputs they generate when run through the algorithms.

The intended outcome of this endeavour is simply to understand the biases and idiosyncrasies of each algorithm. It is apparent purely from inspection of the algorithms that they will expose differing spatial aspects and cannot be directly compared (with the exception of the two PHs). The approach is therefore quite exploratory, and we do not define any 'figures of merit' as done in [13]. There is also no attempt to extract original panning parameters from a mixture since the descriptors we are comparing are inherently based in human perception, and we are seeking to describe the spatial image as a whole rather than measure any specific Direction of Arrival (DOA) information.

3.1. Constructed signals

Two specially constructed signals are used to examine the workings of spatial descriptors under test. The first is a combination of non-overlapping partials that are statically panned to equally-spaced positions across the azimuthal plane. The second takes the same principle, but applies dynamic panning to each partial in the form of a triangle wave at 1 Hz, with lower frequencies panning within a narrower range. All partials have the equal amplitude. Both signals are generated with a duration of 10s at a sample rate $f_s = 44100$ Hz. A visual depiction of each signal is given in figure 1, and throughout the remainder of this paper they will be referred to as simply 'Static' and 'Dynamic'.

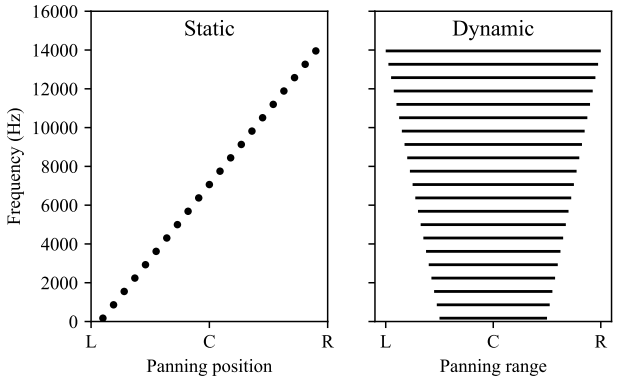


Figure 1: *Two test signals comprised of evenly spaced partials*

The frequencies of the partials are calculated to lie at the centre frequencies of isolated FFT bins, assuring that there are several empty bins in between each one. We define the number of partials $N = 21$, the assumed FFT frame size $F = 2048$, the number of bins separating the partials $S = 32$, and the minimum bin index $M = 8$. The frequencies of the partials are hence given by

$$\begin{aligned} f_n &= \frac{f_s}{F}(nS + M) \\ n &= \{0, 1, 2..., N-1\} \end{aligned} \tag{1}$$

3.1.1. Statically panned partials

The panning value P of each partial is defined to lie in the range $[0, 1]$, where 0 and 1 represent fully left and right respectively. Partials are spaced equally along the azimuthal plane with a buffer B applied at the extreme left and right. The panning values for each partial are given by equation 2.

$$P_n = B + \frac{n(1-2B)}{N-1} \tag{2}$$

For each partial, equal-power amplitude panning is applied to the mono signal. The left and right amplitudes $A_{L,R}$ are calculated by equation 3.

$$\begin{aligned} A_{nL} &= \sin\left(\frac{P_n\pi}{2}\right) \\ A_{nR} &= \cos\left(\frac{P_n\pi}{2}\right) \end{aligned} \tag{3}$$

The final signal is created by synthesizing a sine wave for all partials using the calculated frequencies and amplitudes, and summing them as per equation 4 where T is an integer sample index. A constant gain of $1/N$ is applied to the mixture to prevent clipping, and the phase of each partial ϕ_n is randomized in the range $[0, 2\pi)$ to avoid amplitude spikes during the initial section of the signal.

$$\text{channel}_{\{L,R\}} = \frac{1}{N}\sum_{n=0}^{N-1} A_{n\{L,R\}} \sin\left(\frac{2\pi T f_n}{f_s} + \phi_n\right) \tag{4}$$

3.1.2. Dynamically panned partials

To construct the second signal, we take the first signal as a basis, but replace the constant P_n with a time-varying function - a triangle wave in the range $[0, 1]$, with different partials given different panning intensities and random phases. The panning intensities α_n are given by equation 5.

$$\alpha_n = \frac{n}{2(N-1)} + 0.5 \tag{5}$$

The triangle wave function $\Lambda(t)$ is defined by the equation 6 where $|...|$ is the modulus function, $\lfloor ... \rfloor$ is the floor function, and t is time in multiples of the desired period. The resulting wave will have amplitude in the range $[0, 1]$, completing 1 cycle as t moves from 0 to 1.

$$\Lambda(t) = 2\,|t - \lfloor t + 0.5 \rfloor| \tag{6}$$

The phases of each triangle wave Φ_n are again randomized, but note that $\Phi_n \in [0, 1)$, unlike ϕ_n. The time-varying panning function is therefore given by equation 7.

$$P_n(T) = \alpha_n\left(\Lambda\left(\frac{T}{f_s} + \Phi_n\right) - 0.5\right) + 0.5 \tag{7}$$

The partials are then synthesized using equation 4, replacing the constants $A_{n\{L,R\}}$ with the time-varying $A(T)_{n\{L,R\}}$

$$\begin{aligned} A(T)_{nL} &= \sin\left(P(T)_n * \frac{\pi}{2}\right) \\ A(T)_{nR} &= \cos\left(P(T)_n * \frac{\pi}{2}\right) \end{aligned} \tag{8}$$

The audio files generated using the above method are available at https://github.com/richardmitic/dafx2025.

3.2. Musical excerpts

The excepts listed here for analysis are examples of commercially released music. While the original papers [14, 19] give some examples of how the descriptors react to commercial music, they have been picked specifically to showcase strong amplitude panning. [19] also gives an example of a song that shows low use of amplitude panning, but high de-phasing as a result of multi-tracked vocals. The excerpts here expand on that idea by showing more varieties of spaciousness. They are referred to by the title, artist, year, and timing information. All recordings have a sample rate $f_s = 44100$ Hz.

- *Cut and Clicks*, Tetsu Inoue, 2000, 0:00–0:30. An electronic piece that is characterized by heavy use of extreme panning, unnatural sounds, and singular audio events with very short durations.
- *This Is The Thing*, Fink, 2007, 1:00–1:30. An acoustic song with studio reverb added. The vocal switches from single-tracked to multi-tracked with subtle amplitude panning approximately halfway though the excerpt.
- *Laser Beam*, Low, 2001, 0:00–0:30. Electric guitar, bass, and vocals with an enveloping sound created by reverberation and multi-miking, but very little amplitude panning.
- *Manifesto*, Chilly Gonzales, 2025, 1:00–1:30. Solo upright piano recorded live on stage with an audience present. The spaciousness comes from room-tone and stereo mics placed very close to the piano as opposed to reverb, amplitude panning, or other studio techniques.

4. RESULTS

4.1. Stereo Panning Features

To match the original paper [14], all SPF calculations were performed using a FFT size of 1024, a hop size of 512, and $M = 40$.

It is important to note when interpreting the results of SPF analysis that while the output vector is 16 floating point numbers, some dimensions can be compared against each other while others cannot. There are 4 frequency bands addressed (full, lows, mids, and highs), and the results of the same calculations on different frequency bands *are* directly comparable. However, the 4 summary features comprise combinations of the running mean (μ_r), running standard deviation (σ_r), global mean (μ_g), and global standard deviation (σ_g). The values can only be compared when the 'global' components are the same. We can therefore visualize the results as 2 separate graphs – one for each 'running' feature – with the y-axis denoting value of μ_g and error bars denoting σ_g. Throughout this paper these symbols are combined to reference the SPF features, e.g. μ_r, σ_g to refer to the global standard deviation of the running mean. Figure 2 shows the plots for all test signals and music excerpts.

When comparing the two test signals, we can see several things of note. Firstly, σ_r for the static test signal is always 0, as we'd expect since the panning values never change. Secondly, We can see that for both signals, the amount of panning (measured both by μ_r and σ_r) in highest in the low band, then mid band, and the lowest in the high band. This also makes sense since the signals were constructed with partials spaced equally in the both frequency and panning angle, but the frequency bands are spaced logarithmically. The lower/higher partials are panned generally left/right respectively, meaning that the low band will contain fewer partials

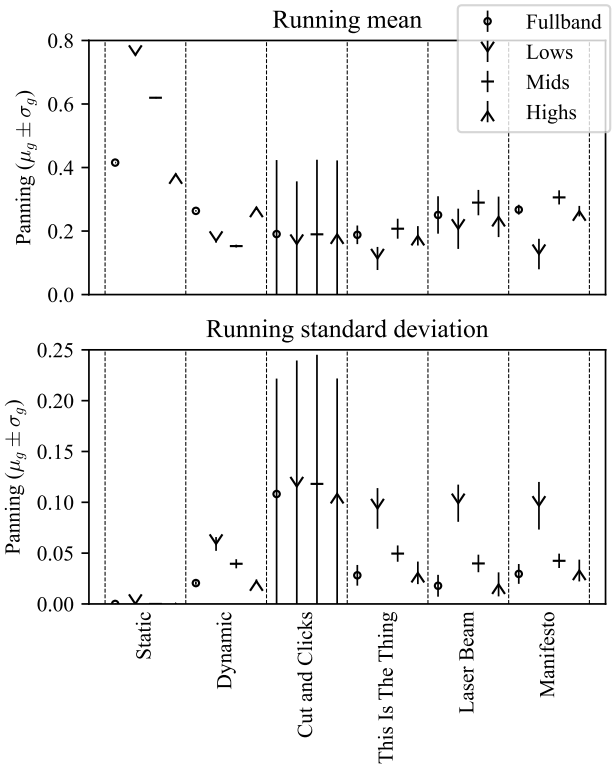


Figure 2: *Stereo Panning Features of test signals and music excerpts.*

panned more left, the mid band will contain more partials panned centrally, and the high band will contain the most partials panned centre and right. Interestingly, we can also see from the graphs that the values of both μ_r and σ_r for the full spectrum band are always very close that of the high band, indicating that (for these signals) the high band is somewhat indicative of the signal as a whole. For both test signals, μ_r, σ_g is always negligible. This makes logical sense because all panning (if any) is performed at an even rate throughout the entire signal. However, σ_r, σ_g is non-negligible, which is a possible indication of a source of measurement error introduced by some component of the algorithm.

Examining the music excerpts in figure 2 we can immediately gather that *Cut And Clicks* contains wider panning variation across all measurements than the other tracks. This makes sense since the track as characterized by only having a single audio event present at one time, panned either completely central or hard left/right, each event lasting for only a fraction of a second. Similarly, the values of μ_r, σ_g for *Manifesto* are lower than all other tracks. This could indicate that this feature is a good indicator of tracks recorded purely with stereo microphones and with very few studio techniques applied. On the contrary, we can see that all the 'natural' tracks (i.e. real instruments recorded with microphones) have very similar values for σ_r, μ_g and σ_r, σ_g. While they are differentiated from the unnatural track, this does call into question whether those measurements are useful for differentiating between natural tracks. Analysis on a much larger dataset would be required to determine which of the 16 dimensions in the SPF are relevant.

4.2. Panning Histograms

All PH measurements were performed with an FFT frame size of 2048, hop size of 1024, and 129 azimuth bins. To compare the two PH algorithms we always display both histograms superimposed on top of each other.

Figure 3 shows the PH plots for the two test signals. The PHs of *Static* show that the two algorithms are each biased towards different azimuth regions. The Gomez algorithm appears to show roughly consistent energy for partials panned near the centre, with energy levels dropping off towards the sides. On the contrary, the Sarroff algorithm exhibits an energy dip in the centre, rising towards the left and right lobes. It is unclear whether either of these phenomena were intentional and there is no source in the literature stating that sounds arriving from any particular azimuth are perceptually louder. We can also observe the effect of the perceptual warping function applied by Gomez by examining the distance between energy spikes in the side lobes of the stereo field. From this analysis we can see that it has the effect of spreading central panning energy out towards the lobes.

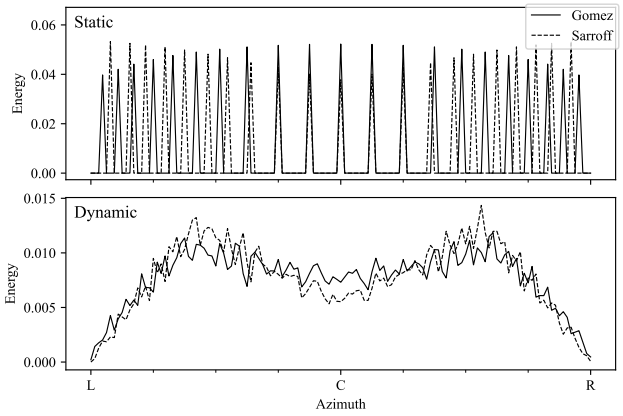


Figure 3: Panning histograms of statically (above) and dynamically (below) panned test signals

Examining the histograms of the dynamically panned signal, firstly we can see that the centre/side biases of each algorithm still apply, although the effect is lessened since each partial is hard-panned for less time overall in the dynamic signal. Most notably, both panning spectrograms give no indication of change over time or differences in frequency bands. [17] does briefly state that the PH can be performed on sub-bands of a panning spectrogram, but doesn't actually demonstrate it.

Figure 4 shows the PHs for all music excerpts. For 'Cut and Clicks' both algorithms produce almost exactly the same histogram, although they make an egregious error; Since the mean over the entire duration of the excerpt is taken, it appears that almost all energy is in the centre, when in fact it is split evenly between hard left, centre, and hard right. This highlights a limitation of the PH as a stereo image descriptor – it does not account for the speed and magnitude of panning variation over the duration of analysis.

Examining the PHs for *This Is The Thing*, *Laser Beam*, and *Manifesto*, we again see that both histogram algorithms are largely in agreement about location of energy in the stereo field, although they exhibit the same biases seen in figure 3; As one would expect, the bias is slightly less obvious when analysing real music

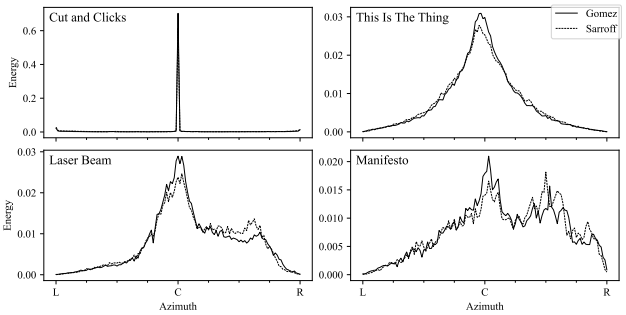


Figure 4: *Panning histograms of music excerpts*

as opposed to artificial signals, but it is nonetheless clearly visible. Analysis of a much larger dataset would be required to determine if it is non-negligible.

Manifesto's PH exhibits a different shape to the other natural tracks – the maximum is lower and energy is more spread across the whole azimuth (note that the y-axes have different scales on all graphs). As previously mentioned, the track was recorded with stereo microphones placed very close to a solo piano. The overall spread of azimuthal energy could therefore be an indicator of this recording style. Moreover, we see that there are several peaks in the right-hand lobe of the histogram. Listening to the excerpt it becomes apparent that these peaks correspond to melodies played on the higher keys (i.e. on the right-hand side of the piano). The musical arrangement is fairly sparse, giving those high melodies room to dominate the stereo field when they are present. This is an indicator that the PHs have not completely disentangled musical content from stereo field or recording style.

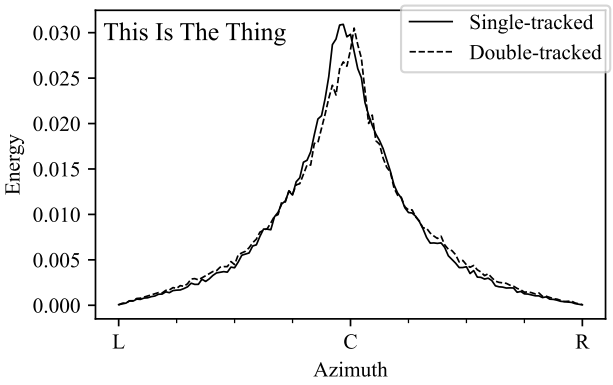


Figure 5: *Panning histograms of This Is The Thing, single and multi-tracked vocals*

The track *This Is The Thing* offers another insight because the vocals switch from single to multi-tracked half-way through. 5 shows the PHs using the Gomez algorithm for each half of the excerpt. There is a slight difference in the location of the central peaks in both PHs, but it is unclear whether this change is large enough to be significant.

Overall, the PH has the benefit of high resolution information in the azimuth domain, but is lacking in temporal information.

4.3. Amplitude/Phase Stereo Features

To inspect the result of ASF analysis we will visualize the mean panning over time for individual frequency bands with the standard deviation shown as error bars, since other all values can be derived from these. All ASF measurements were performed with cochleagrams of 70 sub-bands ranging from 30 to 11050 Hz spaced linearly on the Equivalent Rectangular Bandwidth (ERB) scale. Audio was split into 20ms chunks with a 50% overlap. Figure 6 shows the plots for the two test signals.

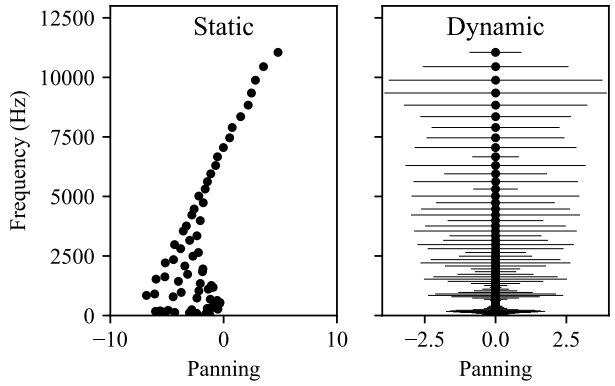


Figure 6: *Average amplitude panning and variation by cochleagram frequency band for test signals*

Firstly, we can see that the static signal shows negligible panning variation, which is expected. The left-to-right trend as frequency rises is also expected, although there is a lot of variation in the low-mid frequencies. This is most likely due to the incongruence between the partial frequencies (which are spaced linearly) and the cochleagram filters (which are spaced logarithmically). In the lower frequencies some cochleagram bands contain no energy at all, while in the higher frequencies, a single cochleagram band contains multiple partials. Similarly, the panning variation in the dynamic test signal appears to have roughly the expected trend in the lower frequencies but tends to be very noisy in the higher frequencies where multiple partials are blurring the panning. In general, this indicates that the ASF is quite an accurate representation of amplitude panning, and the high number of frequency bands adds information that was lacking in SPF.

Figure 7 visualizes the ASF for all music excerpts. Note that unlike figure 6, the frequency scale is logarithmic to more closely match both the cochleagram bands and the musical content. Two tracks present interesting features; *Laser Beam* clearly shows different amplitude panning in different frequencies, which most likely correspond to the bass and guitar in the original recording. *Manifesto* shows fairly high mean panning values across the piano's major frequency range, possibly corresponding to the locations of the piano strings. Interestingly, the panning variation in the piano is far greater than that of the two instruments in *Laser Beam*. The reason for this is unclear, but it is a strong indicator that the two tracks were recorded using different techniques.

The ASF plot for *Cut and Clicks* shows a mix of results similar to both SPF and PH. The average panning position in generally centred, but there is very wide variation in all frequency bands, although slightly less at higher frequencies. The difference between *Cut and Clicks* and the other 'natural' tracks is not as clear as in

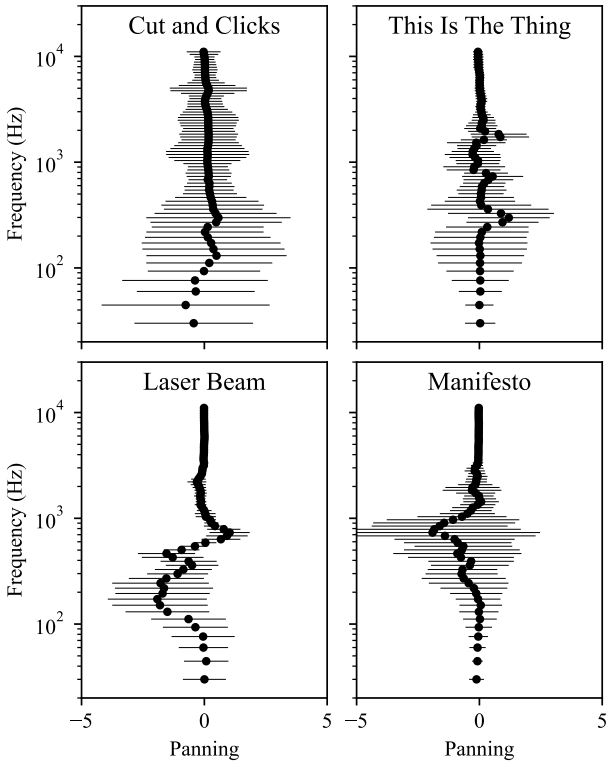


Figure 7: *Panning and variation by cochleagram band for music excerpts*

the SPF plots, although still visible. This indicates that the features of ASF not shown in the plots might still be necessary to provide a full representation of the stereo image.

As mentioned in section 4.2, we can examine the difference between single and multi-tracked vocals by comparing the two halves of *This Is The Thing* in figure 8. This time, the difference is clearly visible as an increase in the panning distribution while the mean panning positions stay roughly equal.

The ASF appears overall to be useful indicator of spatial image, showing detail in both the frequency and time dimensions, although it lacks the high-resolution azimuthal information of the PH.

4.3.1. Issues with Phase Stereo Features

Calculating the PSF presents some challenges. According to [19], the SSPS (denoted by σ_{LR}) for a given TF bin is calculated by equation 9 where t is time, and ω,ϕ are frequency and phase derived via an STFT.

$$\sigma_{LR} = \frac{\sigma(\cos(\omega t + \phi_L) - \cos(\omega t + \phi_R))}{\sigma(\cos(\omega t + \phi_L) + \cos(\omega t + \phi_R))} \tag{9}$$

By inspection, we can see that when $\phi_L - \phi_R = \pi$ (i.e. left and right channels are exactly out of phase), the denominator will equal 0 and σ_{LR} is therefore undefined. Moreover, as $(\phi_L - \phi_R) \to \pi$ the denominator will be very small, and hence σ_{LR} will be very large, climbing to an asymptote at $\phi_L - \phi_R = \pi$. The practical

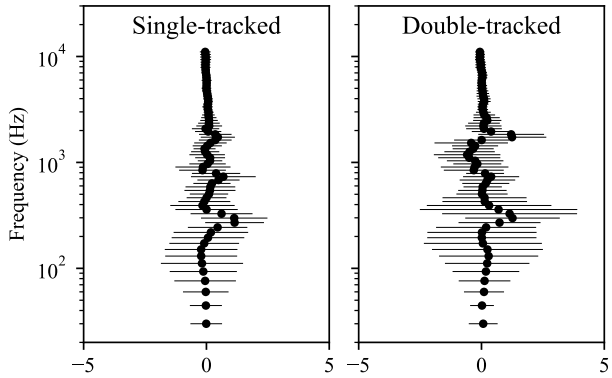


Figure 8: *Panning and variation by cochleagram band for This Is The Thing, single and multi-tracked vocals*

consequence is that a few extremely large values of σ_{LR} are interspersed with a majority of lower values. Since the PSF features are calculated by taking the mean and standard deviation over various axes of the SSPS, these values are prone to errors caused by extremely large outliers having a disproportionate contribution to the mean. [19] does not specify how to deal with these situations. Because of this, we have chosen to omit PSF from the comparison. Nevertheless, it is clear from this and other literature [4] that phase information is important and should not be dismissed in future work.

5. DISCUSSION

All the stereo image descriptors examined have both benefits and drawbacks, usually appearing as trade-offs between the resolution of azimuth, frequency, and time information. Determining a sensible point at which to perform these trade-offs requires consulting the results of relevant perceptual tests, which is beyond the scope of this paper. Another clear trade-off is computational complexity, which was not addressed in this paper but varied strongly between algorithms. STFT-based descriptors ([14, 17]) are efficient but come with and inherent time-frequency trade-off and are sometimes difficult to reconcile with human perception, whereas algorithms based on cochleagrams ([19]) produce more perceptually relevant results but require very large banks of filters.

An additional variable not covered by this paper is the duration of analysis. 30s was chosen to roughly match [14], although [17] mentions that any duration between 2s and the entire recording is acceptable and [19] shows examples of entire songs.

Overall it is apparent that there is no clear consensus in the literature on how such variables should be decided, or how to compare the results of each stereo image descriptor. A typical use case for all descriptors is a classification task where the classes are known a-priori [31, 14, 18, 22, 15, 20, 19, 32, 24], but there has been no survey on how closely any high-resolution descriptors match human perception. In this paper, the use of specially constructed test signals was used a workaround to fill that void, but ideally any further research would be backed by a psychoacoustic model or perceptual tests.

The fact that so much previous research had classification as a use case is a strong sign that the descriptors' values only make sense in relation to a larger population. However, in many cases said population was a private musical collection, which limits the reproducibility of the papers' results. It was also frequent in the practical implementations of the algorithms to hit divide-by-zero errors or other mathematically undetermined phenomena. These are typically not addressed in the original papers.

6. CONCLUSION

The inner workings of four different stereo image descriptors were analysed by feeding their algorithms firstly with signals specifically constructed to provoke extreme responses, and then with excerpts of commercial music picked for their specific flavours of spaciousness. The results show that while all descriptors show good potential in being able to showcase the stereo field of different recordings, there is no one-size-fits-all solution. To gain precise insight in a recording's stereo field, an ideal stereo image descriptor would give a) high-resolution information in the spatial domain with said information comprising both amplitude and phase components, b) indication of changes over time at both subsecond and musical-form resolution, and c) all the above calculated over perceptually relevant frequency bands.

Future work should therefore concentrate on incorporating parts of existing algorithms into a useful stereo image descriptor. Additionally, the inclusion of other spatial descriptors such as those focused on binaural audio and ML might prove fruitful. To that end, a much larger survey of spatial analysis techniques is required. Likewise, analysis of a much larger (and more open) musical dataset would generate better insights than those gained here, and might enable the creation of a much-needed standardized metric for the assessment of spatial descriptors.

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